QUALITY INFORMATION DOCUMENT

For Ocean Health Bloom BALTIC OceanColour

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RECORD TABLE

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1) INTRODUCTION

a) Product Name

OMI_HEALTH_BLOOM_BALTIC_spatiotemporal_coverage

b) Indicator Evaluation Method

The CMEMS Ocean Colour OMIs are derived from the reprocessed (MY) regional gridded chlorophyll products generated and distributed by the CMEMS OC-TAC. The input dataset are the spectral remote sensing reflectances (Rrs) produced by the Plymouth Marine Laboratory (PML) using an ad-hoc configuration for CMEMS of the ESA OC-CCI processor version 6 (OC- CCIv6) to merge at 1km resolution (rather than at 4km as for OC-CCI) MERIS, MODIS-AQUA, SeaWiFS, NPP-VIIRS and OLCI-A data. The chlorophyll product is derived from a Multi-Layer Perceptron neural-net (MLP) developed on field measurements collected within the BiOMaP program of JRC/EC (Zibordi et al., 2011). The algorithm is an ensemble of different MLPs that use Rrs at different wavelengths as input. The processing chain and the techniques used to develop the algorithm are detailed in Brando et al. (2021; 2024).

In the Baltic Sea, the phytoplankton blooms occurring during summer are dominated by nitrogen-fixing cyanobacteria. These blooms can have subsurface and/or surface accumulations, depending on the age of the bloom. Consistent with the HELCOM environmental reporting (e.g. Öberg 2018) the cyanobacterial subsurface and surface summer blooms are detected by applying the thresholds on Rrs at the wavelength of 555 and 670 nm (Hansson and Håkansson 2007). The thresholds applied to daily Rrs images were set at Rrs (555) > 4.25 10−3 sr^{-1} and Rrs (670) > 1.22 10−3 sr^{-1} for the subsurface and surface blooms, respectively, as defined in Kahru et al (2007). This Rrs-based approach to detect cyanobacterial blooms allows overcoming the difficulty of retrieving accurate Chla values in summer bloom conditions (Hansson and Håkansson 2007; Brando et al 2021).

The chlorophyll dataset of the Baltic Sea MY contain the cyanobacteria summer bloom mask variable (CYANOBLOOM), which has the same dimension as the chlorophyll data and reports whether the thresholds for the subsurface and surface blooms was exceeded on the daily Rrs images. As cyanobacterial blooms are extremely patchy and the surface scum is unlikely to totally cover a 1 km resolution pixel, it is possible for both thresholds to be exceeded concurrently.

<u>Product: OMI_HEALTH_BLOOM_BALTIC_spatiotemporal_coverage</u>
This product includes the time series of the total coverage of surface and subsurface blooms in summer for the whole basin on annual basis. Following Hansson and Håkansson (2007), the cyanobacteria summer bloom mask variable (CYANOBLOOM) in the MY regional chlorophyll dataset was aggregated from from June to September (i.e., days 161 to 270) of each year for the three possible values (subsurface =1; surface=2, and 3: both subsurface and surface summer blooms detected). Then the cumulative maps for every year for the three bloom classes were summarized as a spatiotemporal coverage and expressed in day km^2.

c) Copernicus Marine products used as input for the computation

Product name (and Type)	Documentation
OCEANCOLOUR_BAL_BGC_L3_MY_009_134	QUID: https://documentation.marine.copernicus.eu/QUID/CMEMS-OC-QUID-009-131to134.pdf PUM: https://documentation.marine.copernicus.eu/PUM/CMEMS-OC-PUM.pdf

2) SCIENTIFIC VALIDATION

a) Ocean Health Bloom BALTIC OceanColour

The basis of these indicators is the accuracy and stability over time of the source data from the ESA OC-CCI project, as discussed in the QuID for this product: https://documentation.marine.copernicus.eu/QUID/CMEMS-OC-QUID-009-131to134.pdf.

As this method is based on application of thresholds to Rrs images, validation statistics for the spatiotemporal coverage (expressed in day km^2) for the three bloom classes cannot be provided.
The thresholds for the detection of subsurface and surface blooms are applied to daily images of Rrs (555) and Rrs (670): these two CCIv6.0 Rrs bands used as input for the OMI show an adequate agreement within situ data in such optically complex waters in terms of correlation (Determination coefficient 0.78-0.90).
Nevertheless, the use of this Rrs-based approach in the OMI allows overcoming the difficulty of retrieving accurate Chla values in summer bloom conditions (Hansson and Håkansson 2007; Brando et al 2021).

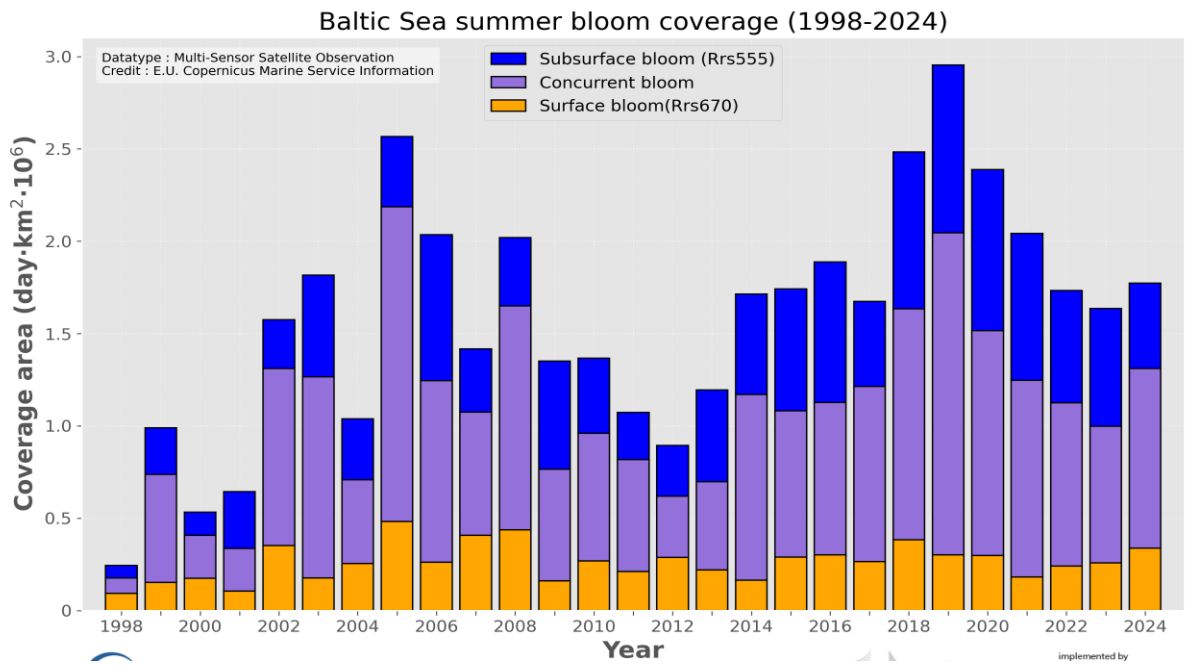

Figure 1: Summer subsurface and surface bloom coverage time series for the Baltic Sea. Concurrent blooms are the areas where surface and subsurface thresholds were exceeded.

3) REFERENCES

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